



MAMOLOTO TECHNOLOGIES (PTY) LTD

#WeServe¹

Company Profile 2018 – 2026



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¹ Our Service Hashtag

MAMOLOTO TECHNOLOGIES (PTY) LTD

Corporate Profile Ver. 3.01 — Solar PV 4.0

Executive Positioning

Mamoloto Technologies (PTY) LTD trading as **Solar PV 4.0** is an **Intelligent Energy Systems Company**, delivering advanced, integrated energy solutions across South Africa.

We engineer systems that go beyond power generation — combining:

- Renewable energy
- Electrical infrastructure
- Intelligence systems
- Digital control layers

We don't ONLY install Solar PV — **We architect intelligent energy ecosystems.**

Vision

*To lead Africa's transition into **intelligent, data-driven energy systems** through Solar PV 4.0 innovation.*

Mission

To design, build, and optimise **future-ready energy systems** that are:

| Reliable | Sustainable | Intelligent |

1. Solar PV 4.0 — Core Identity

Defining the Category

Solar PV has evolved into a new paradigm:

<i>Generation</i>	<i>Capability</i>
<i>PV 1.0</i>	Basic generation
<i>PV 2.0</i>	Grid-tied systems
<i>PV 3.0</i>	Hybrid + storage
<i>PV 4.0</i>	Intelligent, connected, self-optimising energy systems

Mamoloto Solar PV 4.0 Framework

Layer 1: Energy Generation

- Solar PV arrays
- Hybrid inverters
- Battery Energy Storage Systems (BESS)

Layer 2: Electrical Infrastructure

- LV/MV integration
- Protection systems
- Substations & distribution networks

Layer 3: Intelligence Layer (Differentiator)

- Smart metering
- Real-time monitoring
- Load profiling
- Fault diagnostics

Layer 4: Digital Twin & Control Layer

- System simulation models
- Predictive maintenance
- Performance optimisation
- Remote system control

This layer transforms energy systems into **living, data-driven assets**.



A Division of Tlabela Holdings #WeServe

Company Profile

Reliable Energy.
Disciplined Systems.
Long-Term Value.

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Who We Are

Mamoloto Technologies is a South African solar PV services company. We specialise in installation, optimisation and long-term asset management of Solar PV, Battery Storage, Hybrid Energy Solutions and Solar O&M services. Our focus is on ensuring that our energy solutions realise performance, structured maintenance and long-term value for our asset owning residential communities.



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What We Do



Solar PV System
Design & Installation



Battery Energy Storage
Systems (BESS)



Hybrid & Backup Power
Solutions



Solar PV Operations
& Maintenance (O&M)

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Our Operating Philosophy



Solar PV systems are long-term infrastructure

This requires us to ensure that our clients are confident in the long-term value of their investment.



Maintenance is not optional; uptime matters.

Our structured maintenance approach ensures that our clients' systems are always up and running.



The goal is predictable performance and accountability.

Our structured approach ensures that our clients are confident in the long-term value of their investment.



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Markets We Serve



Residential



Small Business



Guesthouses & Lodges



Industrial & Commercial

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Our Geographic Footprint



Gauteng

Pretoria & Johannesburg



North West

Boputha & Mankweng



Limpopo

Mokopane & Gijani

Travel outside standard service zones on request. We are committed to providing the best service possible.



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Why Mamoloto Technologies



Local Expertise



Proven Installations



Structured Maintenance Systems



Financial Discipline



Long-Term Partnerships



After Installation Accountability

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2. Strategic Positioning

Mamoloto Technologies operates as an: **Integrated Energy Systems Architect**

Delivering full lifecycle capability:

<i>Phase</i>	Capability
<i>Design</i>	Engineering & modelling
<i>Build</i>	EPC execution
<i>Operate</i>	Monitoring & optimisation
<i>Evolve</i>	Data-driven system upgrades

3. Core Value Proposition

*“We deliver **Solar PV 4.0 systems** — integrating energy generation, electrical infrastructure, intelligence, and digital control into a unified, optimised platform.”*

4. Service Architecture

A. Solar PV 4.0 Systems (Flagship)

- Solar PV system design & EPC
- Hybrid, Off-grid and Grid-tied systems
- Battery storage integration
- Smart monitoring platforms
- Digital twin-enabled optimisation
- Performance analytics & reporting

B. Electrical Infrastructure (LV/MV)

- Installations up to 33kV+
- Mini substations & transformers
- Network protection & commissioning

C. Smart Grid & Intelligence Systems

- Advanced Metering Infrastructure (AMI)
- Grid monitoring systems

- Load and demand analytics
- Grid stability optimisation

D. ICT & Digital Systems (4IR Integration)

- IoT-enabled monitoring
- SCADA systems
- Remote diagnostics & control

E. Engineering Consulting

- Network planning (LV/MV)
- Energy audits (Energy Performance Certificates)
- Maintenance strategy design
- System modelling & simulation

5. Differentiation

Mamoloto Technologies stands apart through:

1. Solar PV 4.0 Ownership

We define and implement **next-generation energy systems**

2. EPC + Intelligence Integration

Most Companies:

- Either install systems, Or analyse systems

We do both — **fully integrated**

3. African-Optimised Engineering

- Designed for grid instability
- Built for harsh environments
- Economically adaptive

4. Data-Driven Energy Systems

- Real-time visibility
- Predictive maintenance
- Lifecycle optimisation

6. Technical Capability

- LV/MV systems up to **132kV coordination**
- Solar PV + hybrid + BESS systems
- Smart grid deployment
- PLC, SCADA, I-EMS automation systems

7. Sectors Served

- Commercial & Industrial
- Residential estates
- Mining & heavy industry
- Public infrastructure

8. QHSE Philosophy

Mamoloto Technologies operates under a **Zero Harm, High Performance culture**:

- Full compliance with OHSA
- Embedded safety systems
- Engineering-first risk mitigation

Safety is engineered — not inspected.

9. Leadership & Organisational Structure

Led by **Dr. Mamoloto Tlabela (PhD)**:

- 35+ years of technical expertise
- 15+ years of leadership experience
- Specialist in:

| Power systems | Smart grid | Energy infrastructure | System optimisation |



10. Track Record

- Delivered multiple Solar PV and electrical projects
- Installed cumulative 1.5 MW+ systems
- Proven capability across:

| Hybrid systems | Backup systems | C & I installations |

SOLAR PV 4.0

INTELLIGENT • CONNECTED • OPTIMISED • SUSTAINABLE

THE NEXT GENERATION OF ENERGY SYSTEMS

We architect intelligent energy ecosystems that integrate generation, infrastructure, intelligence and digital control for maximum performance.



More than power.
It's intelligence.
It's control.
It's the future.

WHY SOLAR PV 4.0?



MAXIMUM ENERGY
Optimised generation for higher yields and greater returns.



INTELLIGENT CONTROL
Real-time monitoring and automation for smarter decisions.



PREDICT & PREVENT
AI-driven analytics for fault prediction and proactive maintenance.



LOWER LIFECYCLE COSTS
Data-driven optimisation reduces operating costs and extends asset life.



SUSTAINABLE IMPACT
Cleaner energy for a greener tomorrow.

OUR SMART MOTTO

Smart, Simple and
Clean Energy Saving Solutions.

SOLAR PV 4.0 SYSTEM ARCHITECTURE

LAYER 4 DIGITAL TWIN & CONTROL LAYER

Optimise, simulate and control for peak performance.



DIGITAL TWIN MODELLING



PREDICTIVE MAINTENANCE



PERFORMANCE OPTIMISATION



REMOTE CONTROL & AUTOMATION



DATA ANALYTICS & REPORTING

LAYER 3 INTELLIGENCE LAYER

Real-time data and analytics for intelligent decision making.



SMART METERING



REAL-TIME MONITORING



LOAD PROFILING



FAULT DIAGNOSTICS



ALERTS & NOTIFICATIONS

LAYER 2 ELECTRICAL INFRASTRUCTURE LAYER

Reliable infrastructure for safe and efficient power delivery.



LV/MV INTEGRATION



PROTECTION & SWITCHING



SUBSTATIONS & TRANSFORMERS



DISTRIBUTION NETWORKS



POWER QUALITY MANAGEMENT

LAYER 1 ENERGY GENERATION LAYER

Generate and store clean energy efficiently.



SOLAR PV ARRAYS



HYBRID INVERTERS



BATTERY ENERGY STORAGE SYSTEMS (BESS)



ENERGY MANAGEMENT SYSTEM (EMS)



BACKUP / GRID CONNECTION



DIGITAL TWIN OF YOUR ENERGY ECOSYSTEM



CONTINUOUS FEEDBACK LOOP
Data flows across all layers to continuously learn, adapt and optimise performance.

KEY BENEFITS



HIGHER ENERGY YIELD
More power from every ray.



SYSTEM RELIABILITY
Reduced downtime, maximum uptime.



OPERATIONAL EFFICIENCY
Automation and analytics improve efficiency.



COST OPTIMISATION
Lower OPEX, longer asset life.



ENVIRONMENTAL IMPACT
Reduce emissions, protect the planet.

IDEAL FOR



COMMERCIAL & INDUSTRIAL



MINING OPERATIONS



RESIDENTIAL ESTATES



PUBLIC INFRASTRUCTURE



AGRIBUSINESS & FARMS

OUR CAPABILITIES

- ✓ End-to-End EPC Solutions
- ✓ Engineering & System Design
- ✓ Smart Monitoring & Control
- ✓ Maintenance & Optimisation
- ✓ 24/7 Support & Performance Management

THE ENERGY LIFECYCLE - DESIGNED FOR EXCELLENCE



DESIGN
Advanced modelling and system engineering.



BUILD
Quality installation and integration by experts.



OPERATE
Real-time monitoring and intelligent operations.



OPTIMISE
Continuous improvement for maximum return on investment.

POWERING TODAY.
INTELLIGENTLY SHAPING
TOMORROW.

